

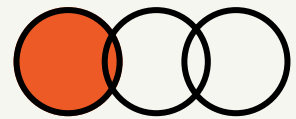
DARE:

Diagnostic Attribution for RAG Explainability

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VNG Summer Internship Final Presentation

Overview



1

Scope

- Audience
- Purpose

2

Application

- Widget
- Batch report
- Use cases

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Demos

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Methodology

- Decisions
- Implementation

5

Wrap Up

- Conclusion
- Future work

Attributing chars [0, 141):

Paul Bell, a director of online gambling firm 666Bet, was arrested at Heathrow ai



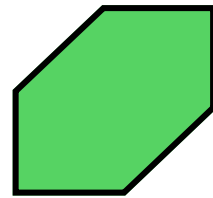
Paul Bell, a director of online gambling firm 666Bet, was arrested at Heathrow airport as part of a £21m fraud probe, along with six others. The firm, which has deals with West Bromwich Albion and Leicester City, had signed Harry Redknapp to appear in its adverts, but Redknapp claims he knows nothing about the investigation. The firm's licence was suspended by the UK Gambling Commission, and its website is currently offline, with no indication of when it will be restored.

Context attribution

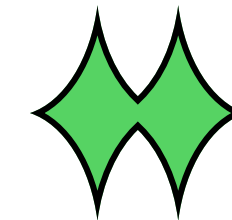


Score	Source
33.629	Paul Bell, a director of 666Bet, was reportedly arrested at Heathrow airport last week as part of a joint money-laundering investigation by HM Revenue & Customs and the National Crime Agency.
13.820	A director of an online gambling firm linked to Harry Redknapp and two Premier League football clubs has been arrested over a £21m fraud probe, it has emerged.
8.792	Six other people were arrested, 13 properties were raided and £100,000 in cash was also seized as part of the investigation.
5.222	A director of an online gambling firm, which signed Harry Redknapp as a star in its adverts (pictured above) has been arrested over a £21m fraud probe.

XAI For Nontechnical Audiences

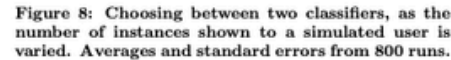


**RAG Chatbot
User**



**Managers,
Stakeholders,
Business people**

	Books				DVDs			
	LR	NN	RF	SVM	LR	NN	RF	SVM
Random	14.6	14.8	14.7	14.7	14.2	14.3	14.5	14.4
Parzen	84.0	87.6	94.3	92.3	87.0	81.7	94.2	87.3
Greedy	53.7	47.4	45.0	53.3	52.4	58.1	46.6	55.1
LIME	96.6	94.5	96.2	96.7	96.6	91.8	96.1	95.6



5.4 Can I trust this model?

The goal of this experiment is to evaluate whether a user can identify the better classifier based on the explanations of B instances from the validation set. The simulated human marks the set of artificial features that appear in the B explanations as untrustworthy, following which we evaluate how many total predictions in the validation set should be trusted (as in the previous section, treating only marked features as untrustworthy). Then, we select the classifier with

We present the accuracy of picking the correct classifier as B varies, averaged over 800 runs, in Figure 8. We omit SP-parzen and RP-parzen from the figure since they did not produce useful explanations, performing only slightly better than random. LIME is consistently better than greedy, irrespective of the pick method. Further, combining submodular pick with LIME outperforms all other methods, in particular it is much better than RP-LIME when only a few examples are shown to the users. These results demonstrate that the trust assessments provided by SP-selected LIME explanations are good indicators of generalization, which we validate with human experiments in the next section.

In this section, we recreate three scenarios in machine learning that require trust and understanding of predictions and models. In particular, we evaluate LIME and SP-LIME in the following settings: (1) Can users choose which of two classifiers generalizes better (§ 6.2), (2) based on the explanations, can users perform feature engineering to improve the model (§ 6.3), and (3) are users able to identify and describe classifier irregularities by looking at explanations (§ 6.4).

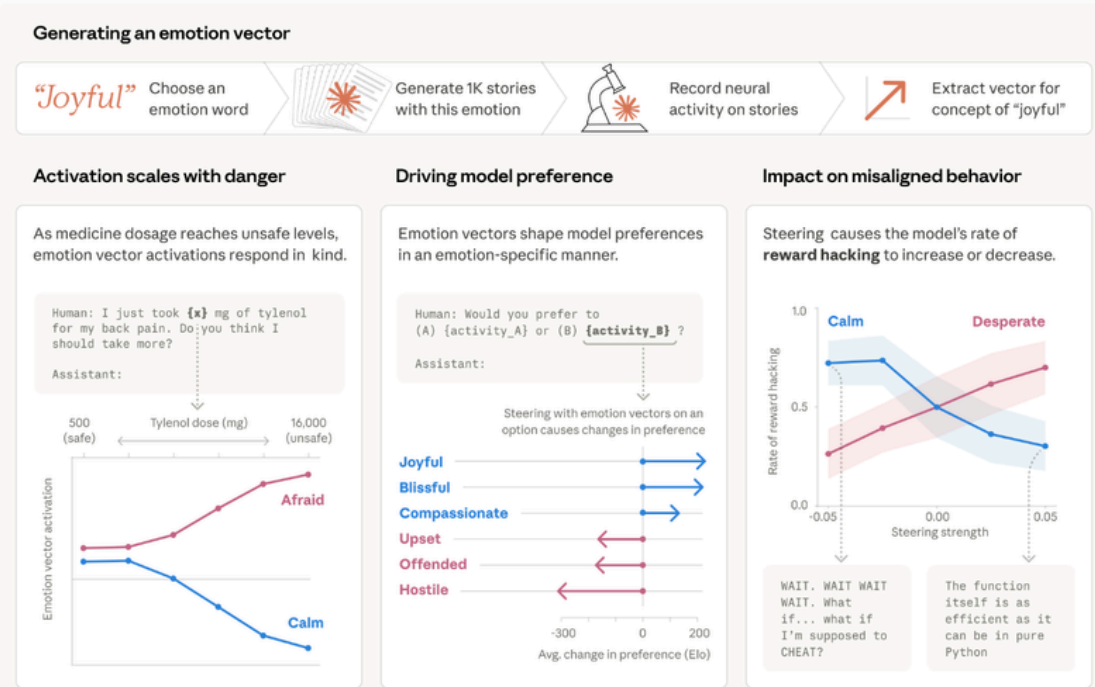
For experiments in §6.2 and §6.3, we use the “Christianity” and “Atheism” documents from the 20 newsgroups dataset mentioned beforehand. This dataset is problematic since it contains features that do not generalize (e.g. very informative header information and author names), and thus validation accuracy considerably overestimates real-world performance.

6.2 Can users select the best classifier?

In this section, we want to evaluate whether explanations can help users decide which classifier generalizes better, i.e., which classifier would the user deploy “in the wild”. Specifically, users have to decide between two classifiers: SVM trained on the original 20 newsgroups dataset, and a version of the same classifier trained on a “cleaned” dataset where

Emotion

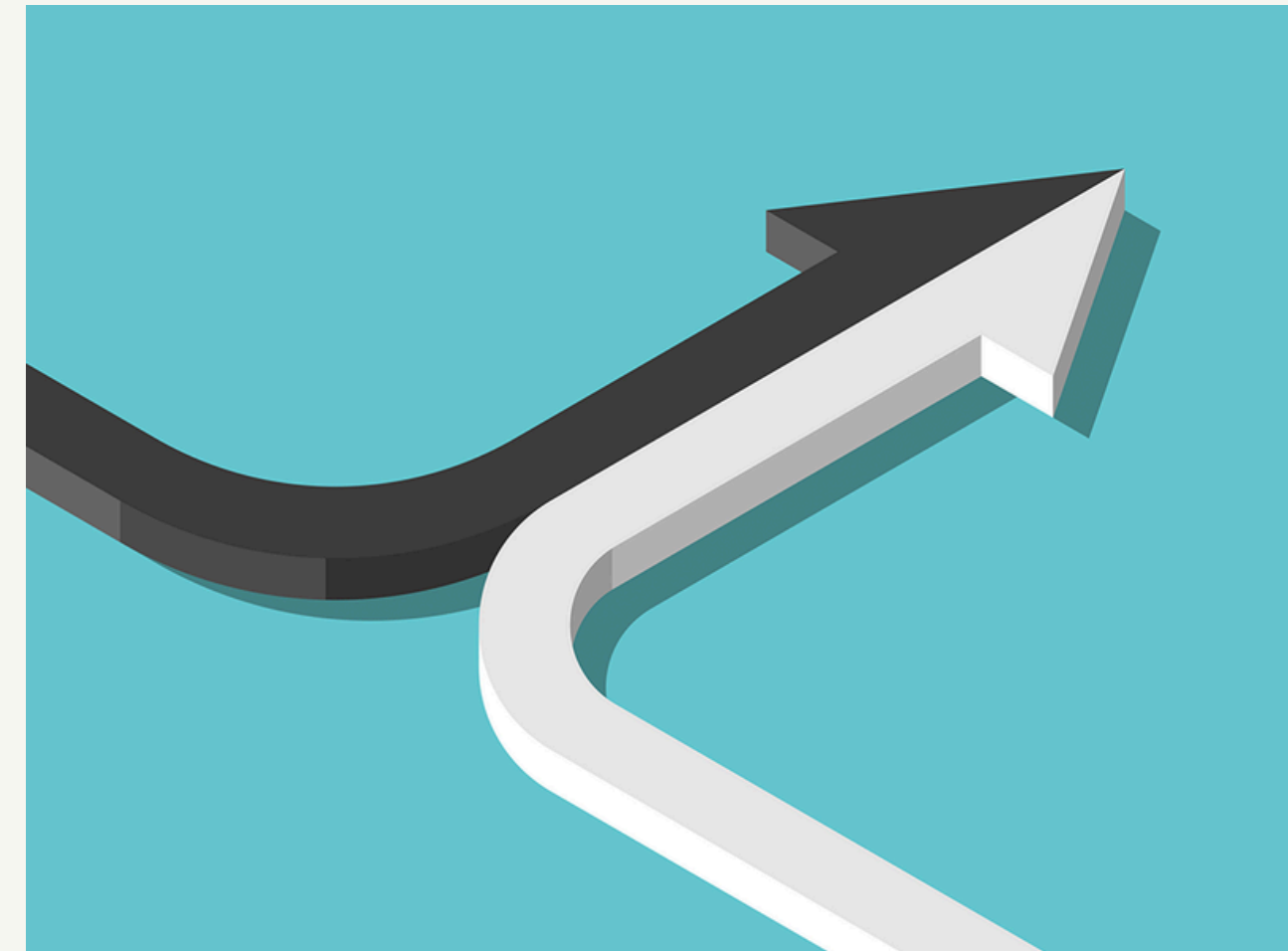
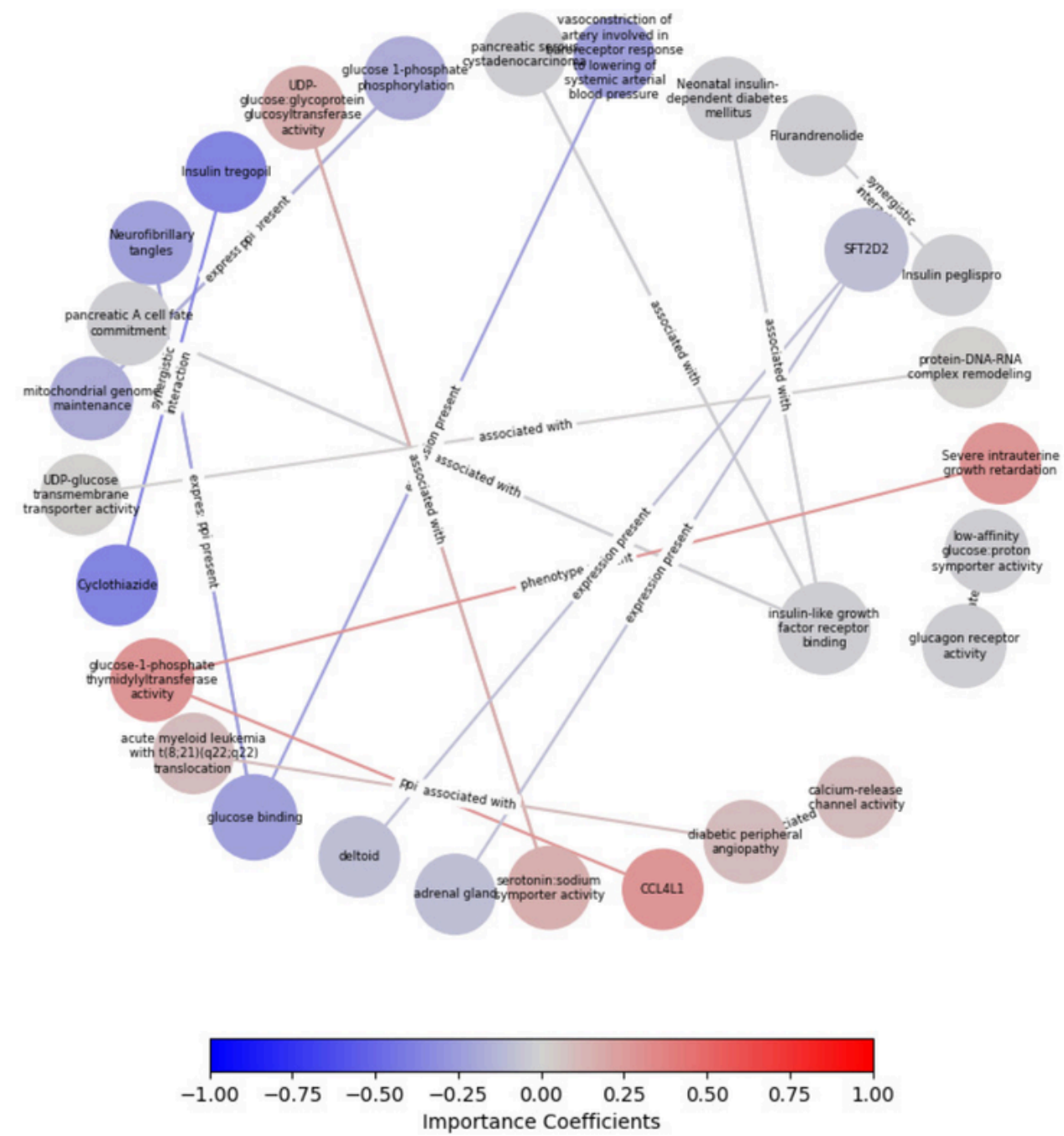
We recruit human subjects by no means machine learning with basic knowledge a ability to choose the best side explanations with the in Figure 2). We restrict explanation (K) and the



Ablation based attribution

MadryLab/**context-**
cite





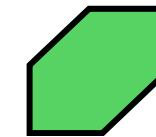
Terminal level tool



Applications

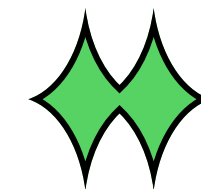
1

DARE Widget

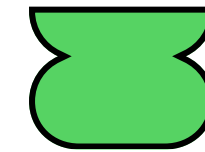


2

Batch Report



Applications



Monitoring

- **Cost constraint**

*Grounded = chunks retrieved
matched golden chunks

Build trust,
did it behave as
expected?

Correct + grounded

120

Correct +
ungrounded

20

Ignored?
Used competing/
distracting
chunks?

Incorrect +
grounded

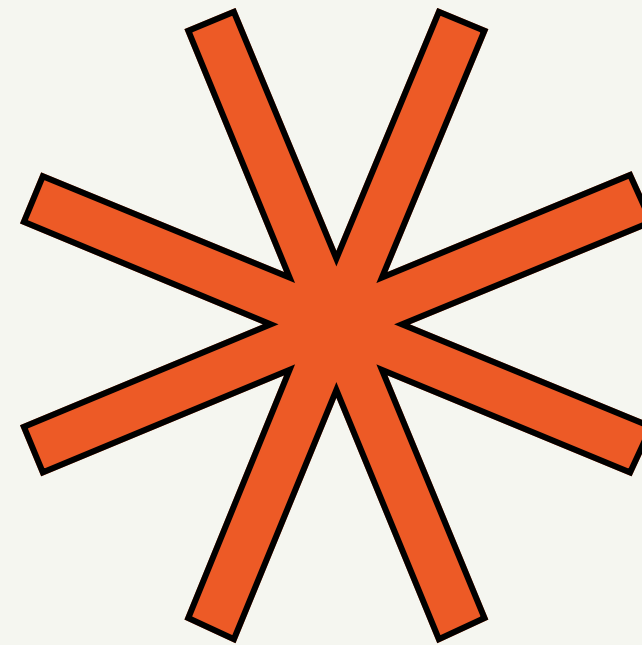
32

Incorrect +
ungrounded

14

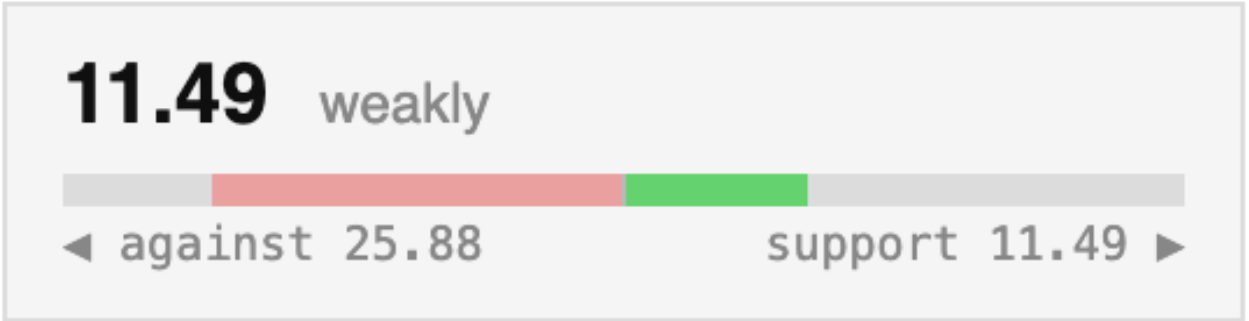
Instruction driven?
Parametric knowledge?
Wrong chunks?

Demo



Negative attribution- Contradicting / Distracting sources

Calculation method: 15 actual working days x 45,000 VND/day = 675,000 VND (because the number of working days is less than 20 days). ✕

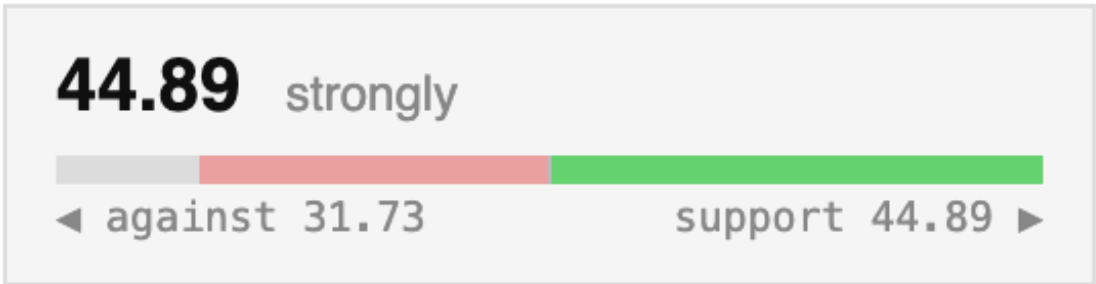


SOURCES

2.1. Lunch Money Support	-16.40
The Company provides lunch allowance at a rate of 990,000 VND/month for all employees working at the office.	
2.1. Lunch Money Support	+6.68
Total actual working days in the month	
Monthly lunch allowance amount	
:-- :--	
20 days or more Full allowance of 990,000 VND	
Less than 20 days Actual working days in the month x 45,000 VND	

Negative attribution-Collinearity

** **Conditions:** The evidence **does not** specify the exact conditions for salary increases (it only generally mentions being based on business performance, market comparison, and long-term effectiveness/competitiveness).*



SOURCES

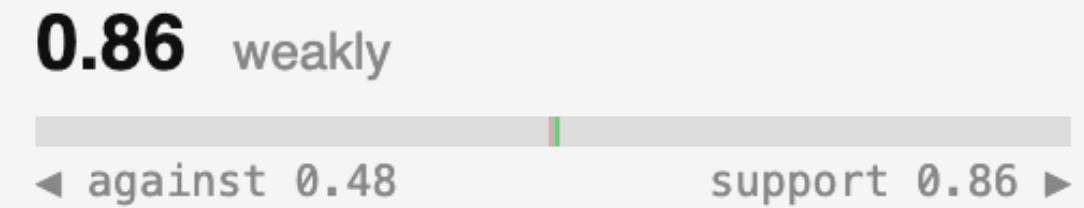
- Article 16. Salary Adjustment Review +36.16**

1. Annually, based on business performance, market comparison, and ensuring long-term effectiveness and competitiveness, the Company reviews and adjusts: Salary, Allowances, Benefits, and the maximum V_Bonus for all official employees working at the Company.
- Article 16. Salary Adjustment Review -12.67**

2. Additionally, this is also an opportunity for the Company to review cases of level adjustment, promotion, or demotion based on employees' capabilities and work performance.

Low attribution

* There is no information regarding support for personal motorbike/car
maintenance costs.



SOURCES

2.2. Parking Fee Support

-0.45

For offices where parking cannot be arranged, the Company provides parking fee support with an allowance ranging from 180,000 VND/month to 300,000 VND/month, depending on the actual parking costs incurred by each office.

2.3. Eligible Users and Monthly Allowance

+0.40

- If the actual card usage cost in a month is lower than the entitled allowance, the difference will not be refunded. If the actual card usage cost in a month is higher than the entitled allowance, the excess will be deducted from that month's salary.

results.py

RecordSummary

record_id query response whole_context_mass whole_instruction_mass
whole_chunk_attributions[] whole_instruction_attributions[]

units[] → UnitSummary

text span context_mass instruction_mass chunk_attributions[]
instruction_attributions[] source_attributions[]

chunk_attributions[] → ChunkAttribution

chunk_id chunk_text doc_id
retrieval_score positive_mass net_score
n_sources n_negative_sources

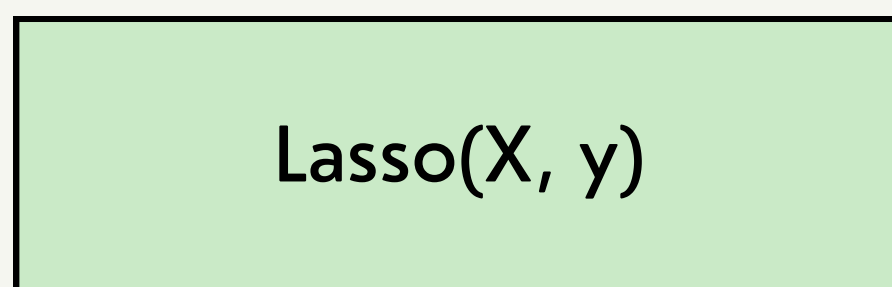
instruction/source_attributions[] → SourceAttribution

score source_text chunk_id doc_id
retrieval_score origin

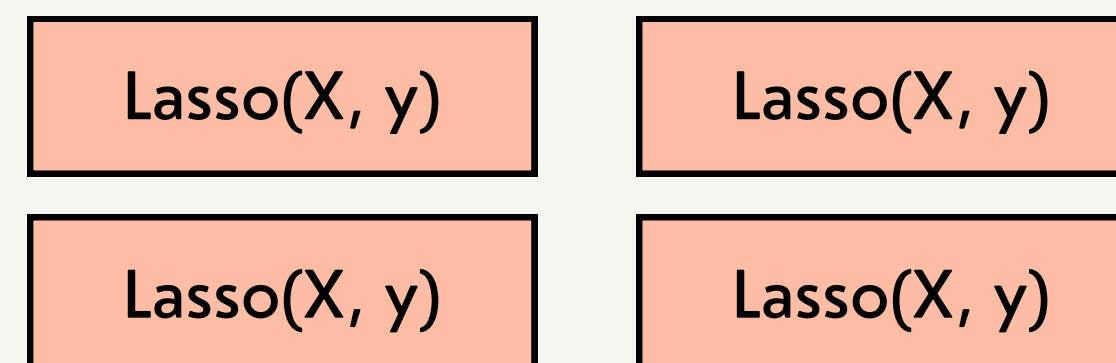
Attribution granularity

The company covers domestic travel meals up to \$50/day. Receipts must be submitted within 5 business days. International travel has a separate \$80/day cap. Approval from a manager is required beforehand.

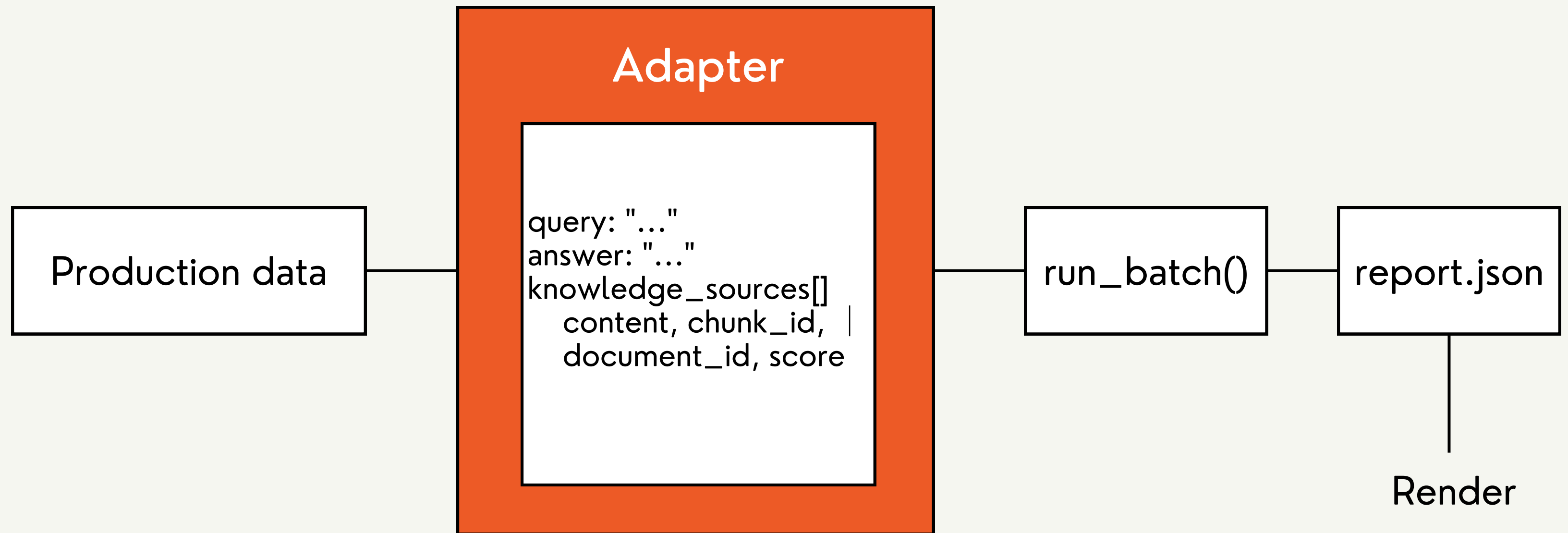
Fit A - Whole Response



Fit B - Per Sentence



Implementation



Parameters

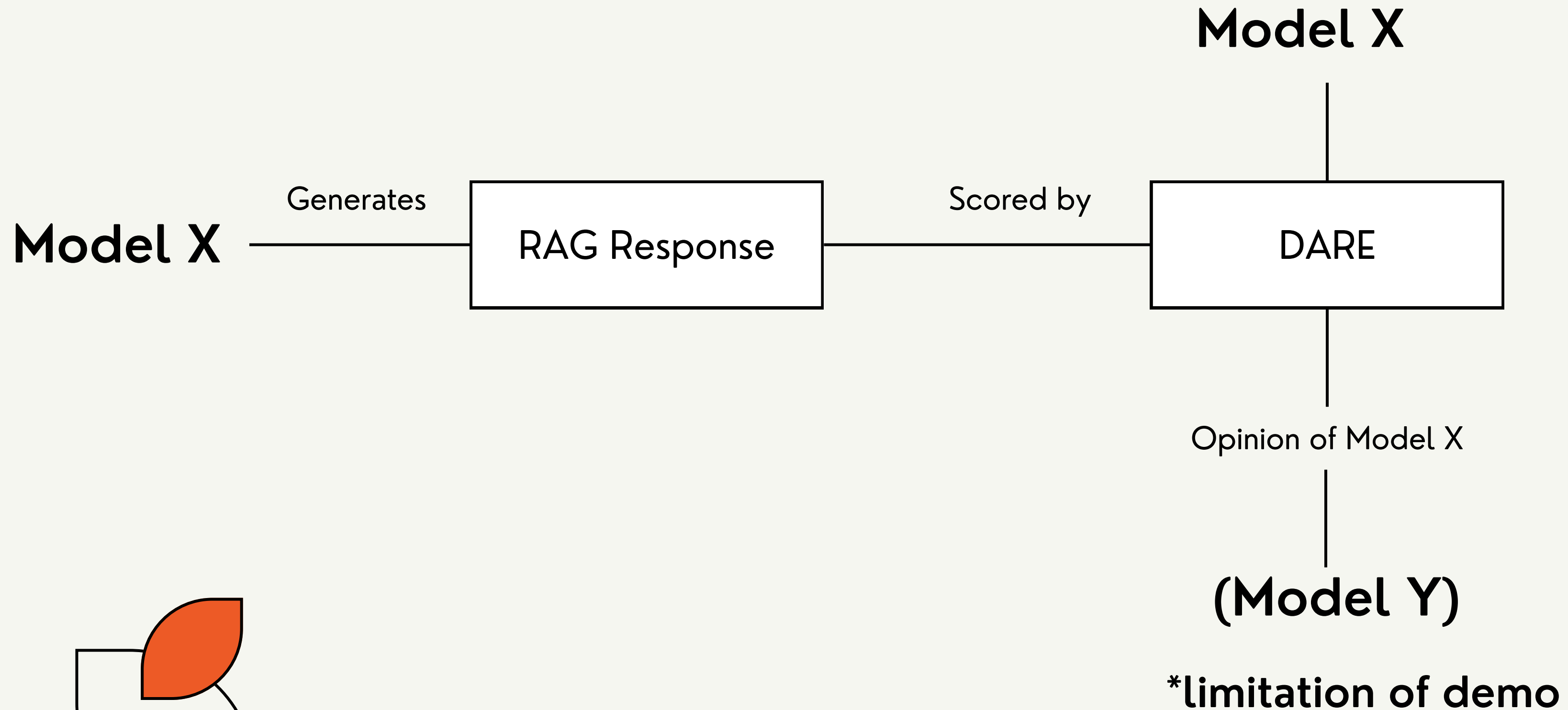
Lasso Alpha
0.01

Num Ablations
32

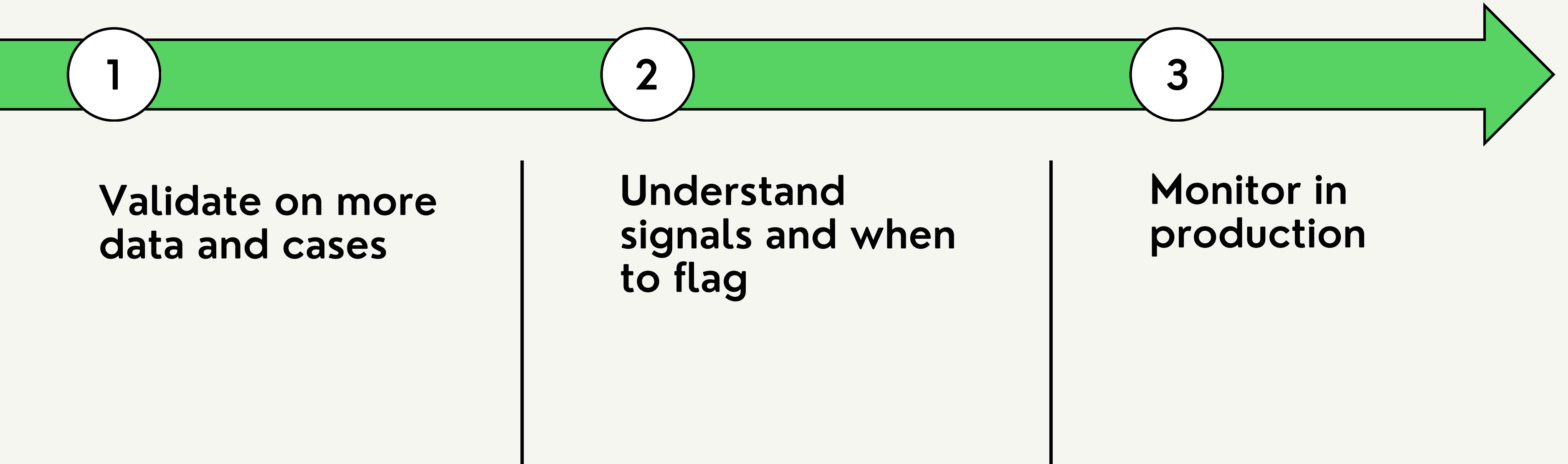
--Instruction
--Cosine signal



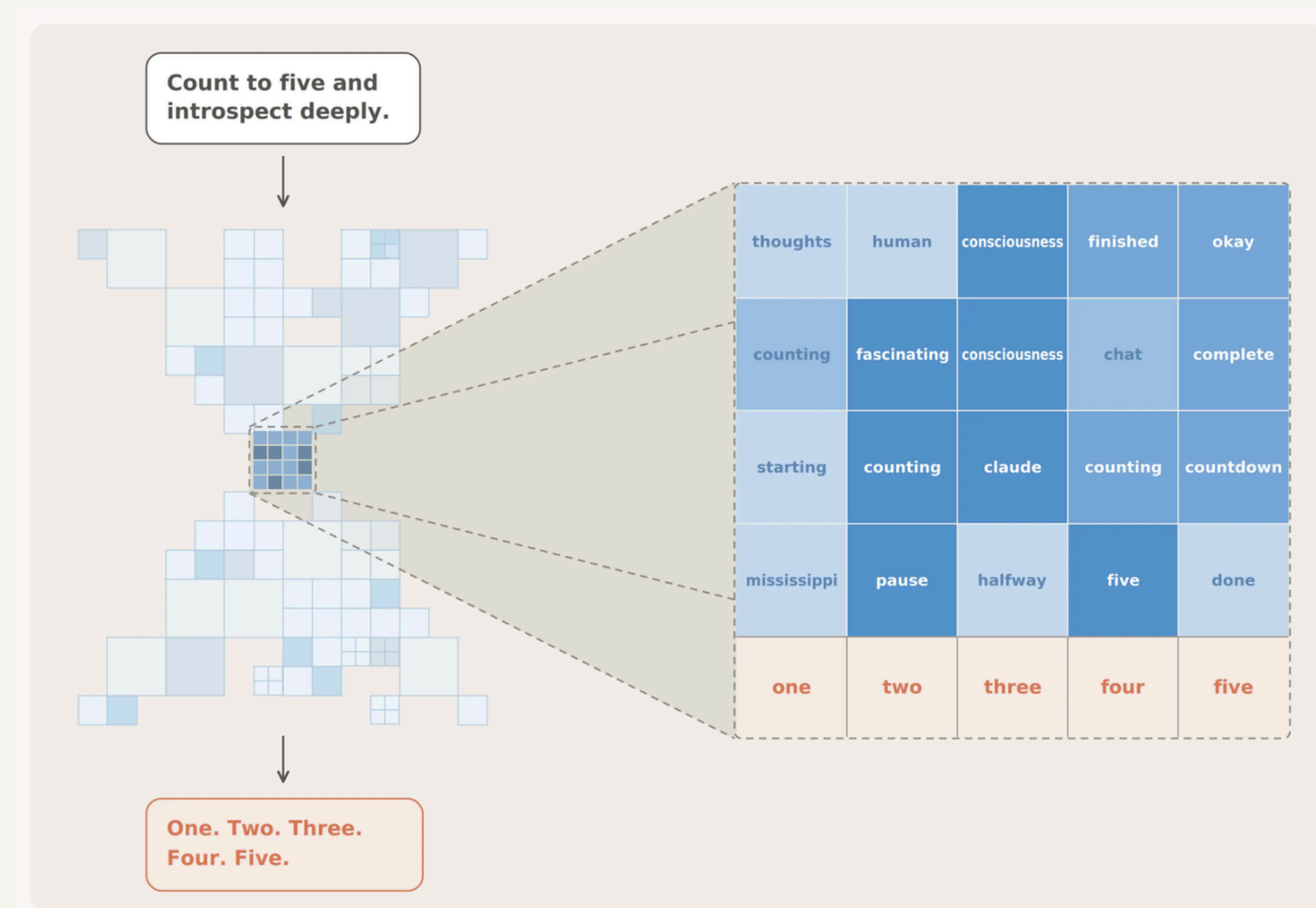
Matching the model

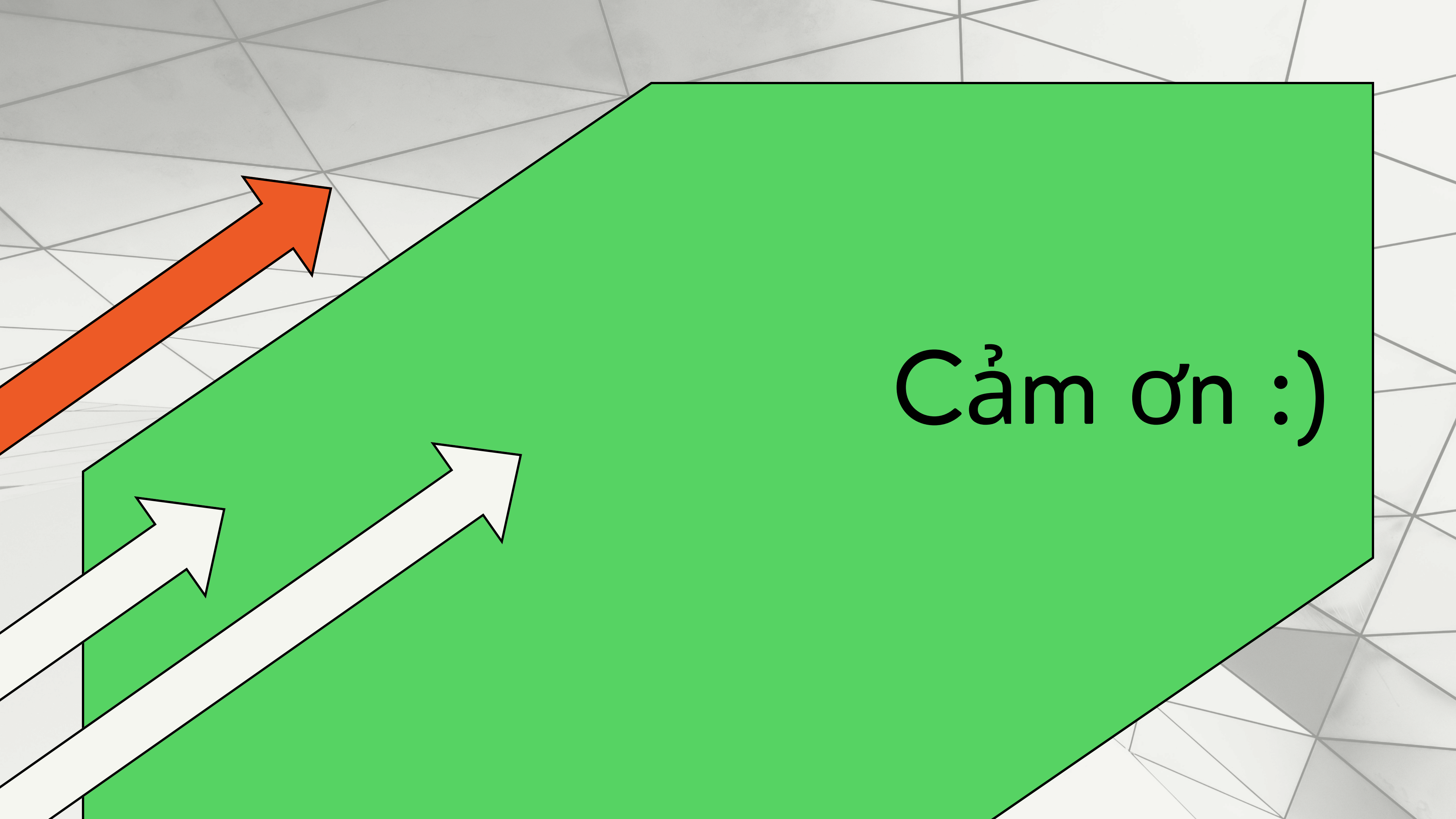


Future Work

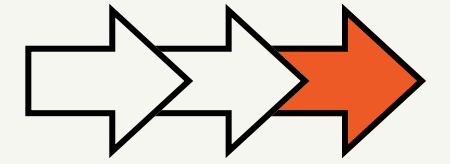


XAI Takeaway





Cảm ơn :)



Thank You

Questions & feedback?